

SET 12: WEBSITE TRAFFIC ANALYSIS

Dataset: Website Traffic

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

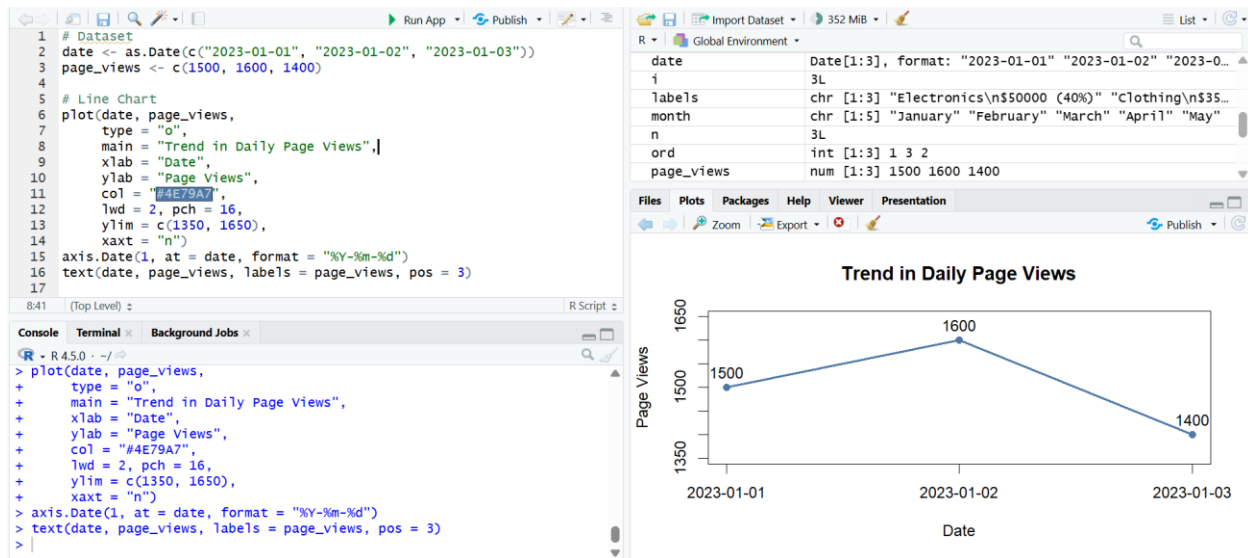
PROGRAM 1: Line Chart of Daily Page Views

Question: Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
# Dataset
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))
page_views <- c(1500, 1600, 1400)

# Line Chart
plot(date, page_views,
      type = "o",
      main = "Trend in Daily Page Views",
      xlab = "Date",
      ylab = "Page Views",
      col = "#4E79A7",
      lwd = 2, pch = 16,
      ylim = c(1350, 1650),
      xaxt = "n")
axis.Date(1, at = date, format = "%Y-%m-%d")
text(date, page_views, labels = page_views, pos = 3)
```

Output 1: Daily Page Views Line Chart



Observation: Page views rose from 1,500 on 2023-01-01 to a peak of 1,600 on 2023-01-02, then dropped to 1,400 on 2023-01-03.

PROGRAM 2: Bar Chart of Top N Days by Click-through Rate

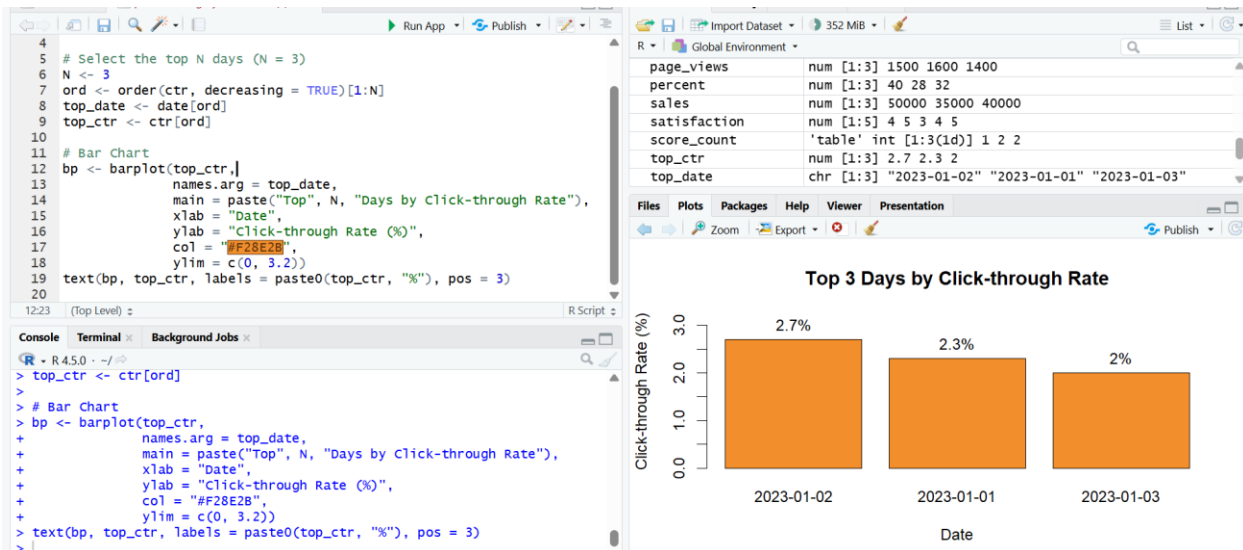
Question: Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Dataset
date <- c("2023-01-01", "2023-01-02", "2023-01-03")
ctr <- c(2.3, 2.7, 2.0)

# Select the top N days (N = 3)
N <- 3
ord <- order(ctr, decreasing = TRUE)[1:N]
top_date <- date[ord]
top_ctr <- ctr[ord]

# Bar Chart
bp <- barplot(top_ctr,
               names.arg = top_date,
               main = paste("Top", N, "Days by Click-through Rate"),
               xlab = "Date",
               ylab = "Click-through Rate (%)",
               col = "#F28E2B",
               ylim = c(0, 3.2))
text(bp, top_ctr, labels = paste0(top_date, "%"), pos = 3)
```

Output 2: Top Days by Click-through Rate Bar Chart



Observation: With $N = 3$, the days are ranked 2023-01-02 (2.7%), 2023-01-01 (2.3%) and 2023-01-03 (2.0%). The day with the most page views also had the highest click-through rate.

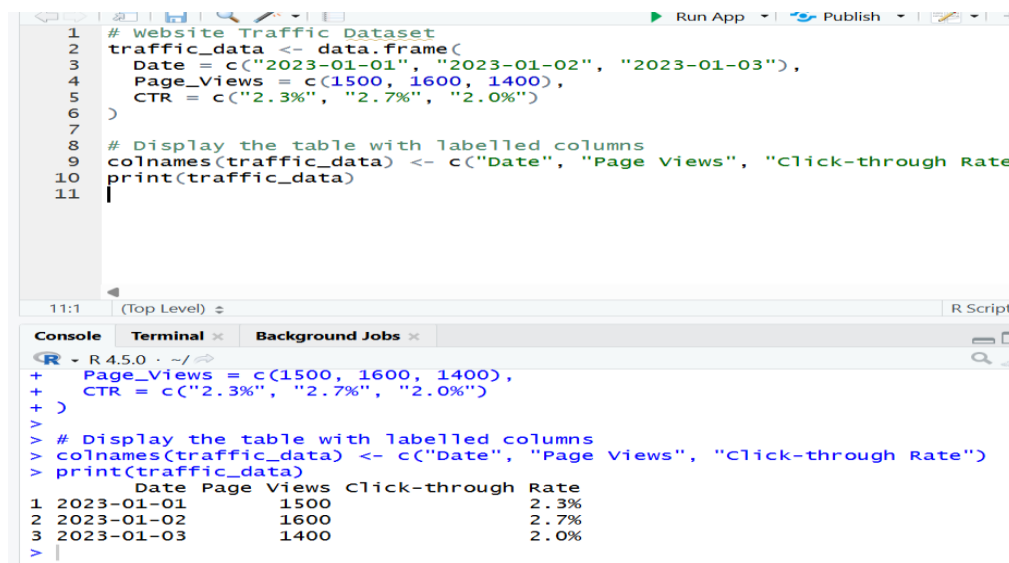
PROGRAM 3: Website Traffic Table

Question: Build a table to show the website traffic data for each day. Label the table elements.

```
# Website Traffic Dataset
traffic_data <- data.frame(
  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),
  Page_Views = c(1500, 1600, 1400),
  CTR = c("2.3%", "2.7%", "2.0%")
)

# Display the table with labelled columns
colnames(traffic_data) <- c("Date", "Page Views", "Click-through Rate")
print(traffic_data)
```

Output 3: Website Traffic Table



```
1 # Website Traffic Dataset
2 traffic_data <- data.frame(
3   Date = c("2023-01-01", "2023-01-02", "2023-01-03"),
4   Page_Views = c(1500, 1600, 1400),
5   CTR = c("2.3%", "2.7%", "2.0%")
6 )
7
8 # Display the table with labelled columns
9 colnames(traffic_data) <- c("Date", "Page Views", "Click-through Rate")
10 print(traffic_data)
11
```

Console Output:

```
> Page_Views = c(1500, 1600, 1400),
+ CTR = c("2.3%", "2.7%", "2.0%")
+ )
>
> # Display the table with labelled columns
> colnames(traffic_data) <- c("Date", "Page Views", "Click-through Rate")
> print(traffic_data)
  Date Page Views Click-through Rate
1 2023-01-01      1500          2.3%
2 2023-01-02      1600          2.7%
3 2023-01-03      1400          2.0%
```

Observation: The table displays the Date, Page Views and Click-through Rate for each of the three days.

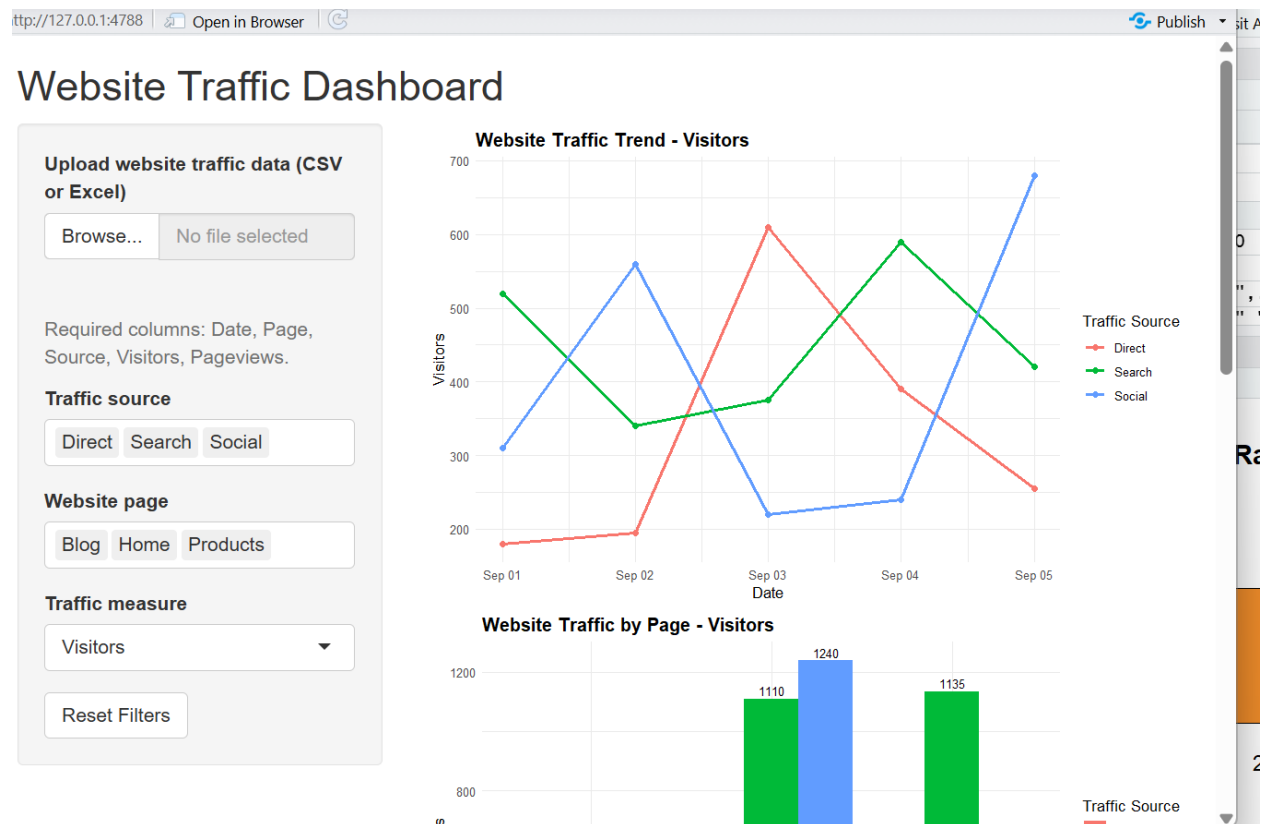
PROGRAM 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

Tableau Steps

1. Open Tableau and connect to the Website Traffic dataset (Excel/CSV).
2. Verify the fields Date, Page Views and Click-through Rate; make sure Date is a Date type and Click-through Rate is a number (convert 2.3% to 0.023 and format as percentage if needed).
3. Create a worksheet named "Page Views Trend". Drag Date to Columns (set to Exact Date) and Page Views to Rows; select Line in the Marks card.
4. Add the title "Trend in Daily Page Views", label the axes (Date, Page Views) and show mark labels.
5. Create a worksheet named "Top CTR Days". Drag Date to Columns and Click-through Rate to Rows; select Bar in the Marks card.
6. Sort the bars in descending order of Click-through Rate. To limit to the top N days, right-click Date > Filter > Top tab > By Field > Top, and enter N (or use a Top N parameter).
7. Add the title "Top Days by Click-through Rate" and show mark labels.
8. Create a worksheet named "Traffic Table". Drag Date to Rows and add Page Views and Click-through Rate as measures (Measure Names/Measure Values) to show the table.
9. Create a new Dashboard, set the size to Automatic, and drag in the three worksheets.
10. Add the title "Website Traffic Dashboard".
11. Select the line chart and click the "Use as Filter" icon; add a Date quick filter (Show Filter) so selections update all views.
12. Test the dashboard by selecting a date or range and checking that the bar chart and table update.

Output 4: Website Traffic Dashboard



Observation: The Tableau dashboard combines the page views line chart, the top click-through rate bar chart and the traffic table, allowing interactive exploration of website traffic through filters and click-to-highlight actions.

Procedure

1. Create the website traffic data (Date, Page Views, Click-through Rate) in R.
2. Convert the dates using `as.Date()` and draw the line chart of page views using `plot()` with `type = "o"`.
3. Sort the click-through rates with `order()`, select the top N days, and draw the bar chart using `barplot()`.
4. Create a data frame containing Date, Page Views and Click-through Rate with labelled columns and display it as a table.
5. Open Tableau and import the Website Traffic dataset.
6. Create the line chart, bar chart (Top N filter) and traffic table as separate worksheets.
7. Combine the worksheets into a dashboard, add a title, and enable filters.
8. Test the dashboard interactivity.

Result

The Website Traffic data was successfully visualized using a line chart of daily page views, a bar chart of the top days by click-through rate, a labelled traffic table, and an interactive Tableau dashboard.